



Project Design Document

1. Project

Aurora Forecast: When Are the Best Conditions to See the Northern Lights?

Aurora Forecast is an end-to-end decision support application that integrates Machine Learning with multiple real-time and forecast APIs to provide current geomagnetic estimates and estimate the probability of observing the Northern Lights for a selected location and date.

2. Problem Statement

People who want to see the Northern Lights often need to consult multiple sources of information, including space weather, geomagnetic activity, weather forecasts and darkness conditions. This process is time-consuming and difficult for non-experts.

The goal of this project is to simplify that process by providing a single application that estimates the probability of observing the Northern Lights for a selected location and forecast date.

3. Project Goal

Develop an end-to-end Machine Learning project that predicts current geomagnetic storm intensity using real-time space weather data and integrates this prediction into a Streamlit application that also provides personalized aurora forecasts for a selected location and date using space weather and environmental API data.

This project builds on my previous Machine Learning project, which focused on predicting whether a Coronal Mass Ejection (CME) would become geoeffective. Instead of modelling the CME itself, this project predicts the resulting geomagnetic storm intensity (Dst index) using current space weather conditions. This real-time prediction is presented as Today's AI Aurora Estimate, while forecasts for selected locations and dates are calculated separately using forecast geomagnetic activity, location, darkness and atmospheric conditions.

The initial idea was to combine the Machine Learning model directly with forecast APIs to generate predictions for future dates. However, the specific solar wind and magnetic field variables required by the trained model are difficult to obtain as forecast data, particularly across longer time horizons. For this reason, the final application uses the Machine Learning model with real-time NOAA measurements, while future forecasts rely on the NOAA 45-Day Ap forecast together with location, darkness and atmospheric conditions.

4. Target Users

- Travelers
 - Aurora hunters
 - Astronomy enthusiasts
 - Anyone planning to observe the Northern Lights
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5. Main Research Question

Can data help users identify the best conditions to observe the Northern Lights?

6. Data Sources

The project uses three historical datasets from NASA's Space Weather collection:

- **Solar Wind:** Minute-level measurements of solar wind conditions (e.g., speed, density, temperature and magnetic field components), used as the main input features.
- **Labels:** Hourly Dst (Disturbance Storm Time) index values, used as the target variable for the Machine Learning model.
- **Sunspots:** Monthly smoothed sunspot numbers, providing long-term solar activity context.

The final integrated dataset supports:

- Data cleaning
- Exploratory Data Analysis
- Machine Learning
- Model comparison
- Model evaluation

Real-Time APIs

- **NOAA Real-Time Space Weather Data** — Solar wind and magnetic field measurements used by the ML model.
- **NOAA Solar Cycle Forecast** — Smoothed sunspot number required by the ML model.
- **NOAA 45-Day Forecast** — Ap and F10.7 forecasts for selected future dates.
- **Open-Meteo Forecast API** — Cloud cover and visibility for dates within the forecast horizon.

- **Open-Meteo Historical Weather API** — Typical historical cloud cover for longer-range dates.
- **Open-Meteo Geocoding API** — Converts the selected destination into latitude and longitude.
- **Sunrise-Sunset API** — Astronomical twilight information used to estimate sky darkness.
- **NOAA Aurora Forecast** — Auroral oval forecast for the next ~30–40 minutes.

Purpose:

➡ Generate personalized aurora forecasts for the selected location and available forecast horizon.

7. Exploratory Data Analysis (EDA)

Objectives:

- Dataset overview
 - Disturbance Storm Time (Dst) Index Distribution
 - Feature distributions
 - Correlation analysis
 - Feature relationships
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8. Data Preprocessing

- Missing value handling
 - Feature selection
 - Train-test split
 - Feature scaling (StandardScaler)
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9. Machine Learning

- Model Training
 - Model Evaluation
 - Model Comparison
 - Feature Importance
 - Hyperparameter Tuning
 - Final Model Selection
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10. Streamlit Application

The application will function as an **Aurora Forecast**, similar to a weather forecast application.

The user will:

- Enter a destination.
 - Select a forecast date up to 45 days ahead.
 - Retrieve space weather and environmental data through APIs.
 - Receive an Estimated Observation Chance for the selected destination and date.
 - View the forecast conditions influencing the estimate.
 - View Today's AI Aurora Estimate based on the Dst index predicted by the Machine Learning model using real-time NOAA data.
 - Explore NOAA's auroral oval forecast for the next ~30–40 minutes.
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11. Streamlit App Workflow

Divided into 3 different parts: **Today's AI Aurora Estimate, Estimated Observation Chance, Interactive Map**

Historical Dataset



Data Cleaning



EDA



Preprocessing



Machine Learning



Best Machine Learning Model



Real-Time NOAA Data



Predict Current Dst



Today's AI Aurora Estimate

Selected Location + Date



NOAA 45-Day Forecast

+ Weather Conditions

+ Darkness

+ Geolocation



Estimated Observation Chance

NOAA Auroral Oval



Interactive Map

13. Technologies

- Python
 - Pandas
 - NumPy
 - Scikit-learn
 - Matplotlib
 - Seaborn
 - Streamlit
 - Requests
 - Joblib
 - Plotly
 - Pydeck
 - APIs
 - **NOAA Space Weather APIs**
 - **NOAA Solar Cycle Forecast API**
 - **NOAA 45-Day Space Weather Forecast**
 - **Open-Meteo Forecast & Historical Weather APIs**
 - **Open-Meteo Geocoding API**
 - **Sunrise-Sunset API**
 - GitHub
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14. Minimum Viable Product (MVP)

- Historical dataset analysis
- Data cleaning and integration
- Exploratory Data Analysis (EDA)
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- Feature engineering and preprocessing
- Machine Learning regression model
- Model evaluation, comparison and tuning
- Real-time API integration
- Streamlit application
- **Today's AI Aurora Estimate**
- **Personalized aurora forecast for selected location and date**
- **Interactive NOAA auroral oval visualization**

15. Success Criteria

The project will be considered successful if it:

- Demonstrates a complete end-to-end data science workflow.
 - Produces a reliable Machine Learning model.
 - Successfully integrates real-time APIs.
 - Provides an intuitive Streamlit application.
 - Allows users to easily understand their estimated chance of observing the Northern Lights.
 - Demonstrates that data can help users identify favourable conditions for observing the Northern Lights.
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16. Future Improvements

- Hourly aurora intensity forecasts for selected future dates.
- Push notifications.
- Travel recommendations.
- Explainable AI.
- Explore approaches for extending the Machine Learning model from real-time estimation to future aurora forecasting.